Divya Shah

divya.shah@savemymails.com |+1 (628) 243-1377| LinkedIn | GitHub | Portfolio | USA (Open to Relocate)

PROFESSIONAL SUMMARY

Senior Software Engineer with 4.5 years of experience building **AI-enabled, cloud-native systems** across finance and retail. Skilled in **Java, Spring Boot, React, Node.js, microservices, and event-driven architectures**. Delivered **credit risk pipelines with RAG and LangChain** enabling analysts to query **4M+ regulatory documents** and implemented cloud storage optimizations reducing costs by **$70K annually**. Experienced in deploying **production-grade applications on AWS with Docker, Kubernetes, and Terraform** and building internal dashboards to improve operational efficiency and insight traceability.

SKILLS

Programming Languages: Java, Python, TypeScript, JavaScript, SQL, C++
Backend & Distributed Systems: Spring Boot, Microservices, RESTful APIs, GraphQL, Hibernate, JPA, Apache Kafka, RabbitMQ, Event-Driven Architecture, Concurrent & Multithreaded Programming, Distributed Systems, Node.js, Express.js, C#
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, DynamoDB, API Gateway), Docker, Kubernetes, Terraform, Jenkins, CI/CD Pipelines, Infrastructure as Code (IaC), Observability (Prometheus, Grafana)
Databases & Data Stores: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch
Frontend Technologies: React.js, Angular, Next.js, Redux Toolkit, Tailwind CSS, Material UI, HTML5, CSS3, SASS, Responsive UI Development
Testing & Quality Engineering: JUnit, Mockito, Cypress, Selenium, Jest, Chai, Postman, Test-Driven Development (TDD)
System Design & Engineering Fundamentals: System Design, High-Level & Low-Level Design (HLD/LLD), Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Design Patterns, Scalability, Performance Optimization
AI & Applied Machine Learning: Generative AI, LangChain, Hugging Face Transformers, GPT, BERT, MLflow, Retrieval-Augmented Generation (RAG)
Security & Development Practices: OAuth 2.0, JWT, Secure API Design, Git, Jira, VS Code, Webpack, Agile/Scrum

PROFESSIONAL EXPERIENCE

JP Morgan Chase &Co. **Jan 2025 – Present | United States**
Senior Software Engineer

- Contributed to the design and enhancement of a **Java Spring Boot**–based distributed platform supporting enterprise credit risk and operational resiliency use cases, integrating data from multiple upstream banking systems and internal risk engines.
- Designed modular **system architectures** for risk evaluation and aggregation services, applying core system design principles such as service isolation, asynchronous workflows, idempotent APIs, and graceful degradation under partial system failures.
- Implemented thread-safe processing components using **Java concurrency utilities** and controlled execution pools to support parallel risk evaluations across **15–20 independent business workflows** while maintaining strict data consistency guarantees.
- Applied **data structures** and **algorithmic techniques** including hash-based indexing and time-window aggregation to optimize lookup and ordering logic across **millions of reference records** used in internal risk assessment pipelines.
- Built and integrated Retrieval-Augmented Generation (**RAG**) workflows using **LangChain** and **OpenAI APIs**, enabling analysts to query **4M+ regulatory documents**, internal policies, and anonymized credit artifacts with source-attributed responses.
- Provisioned and maintained cloud infrastructure using **Terraform**, authoring reusable IaC modules to support **40+ backend services** across **Kubernetes clusters**, IAM configurations, networking, and managed data stores.
- Developed **React-based dashboards** visualizing risk indicators, workflow states, and model outputs for **200+ internal analysts**, including interactive charts, filters, and drill-downs, improving insight traceability and analyst efficiency.

KPIT **Jan 2020 – Aug 2023 | India**
Software Engineer

- Constructed **Spring Boot**–based microservices for a retail order management and inventory platform, supporting **600K–900K daily order and inventory events** across product catalog, pricing, promotions, and fulfillment systems, ensuring stable performance during seasonal traffic spikes.
- Engineered distributed backend services using **RESTful APIs** and **Kafka**, enabling asynchronous processing of order creation, inventory reservations, and shipment updates across **30+ warehouses** and regional retail partners, improving data consistency and system resilience.
- Led database migration from **MySQL** to **PostgreSQL**, restructuring schemas, introducing table partitioning, read replicas, and optimized indexing strategies, resulting in **35% improvement in query performance** and more predictable reporting workloads.
- Managed **Redis** caching for high-traffic retail endpoints (product listings, pricing validation, cart calculations), significantly reducing database load and maintaining **sub-150ms API response times** during peak shopping hours.
- Architected **Angular-based dashboards** for retail operations, including inventory visibility, order tracking, and returns management, actively used by **250+ supply chain, operations, and customer support users** on a daily basis.
- Established **OAuth 2.0** and **JWT-based authentication** across backend services, enforcing role-based access control for internal tools and partner integrations while aligning with enterprise security standards.
- Optimized order processing and fulfillment workflows using **Kafka-based event pipelines**, reducing order failures and manual intervention by **20%** and improving end-to-end order processing reliability.
- Deployed **AWS** services (**EC2, Lambda, S3**) for invoice storage, order artifacts, and audit logs, applying encryption at rest and lifecycle policies, which reduced infrastructure and storage costs by approximately **$70K annually**.
- Configured observability using **Prometheus** and **Grafana**, identifying slow endpoints and resource bottlenecks early, helping maintain **99.8% service uptime** across **18+ microservices**.
- Improved release quality through comprehensive unit, integration, and load testing using **JUnit, Mockito, and JMeter**, reducing post-release defects by **30%** and improving deployment confidence.
- Documented **RESTful and GraphQL APIs** using **Swagger/OpenAPI**, accelerating onboarding and simplifying integrations with third-party logistics and retail partners.

ACADEMIC PROJECTS

AI-Assisted Banking Fraud Monitoring System |Tech: *Java Spring Boot, Spring Security, PostgreSQL, GPT API*

- Programmed a secure transaction monitoring system using **Java Spring Boot** that processed simulated banking transactions for **3,000+ customer accounts**, applying rule-based risk checks to detect anomalous activity.
- Integrated a **GPT-based service** to generate **human-readable risk explanations** for flagged transactions, reducing fraud analyst investigation time by **40%** through faster decision support.

Retail Inventory & Order Management Platform | Tech: *Java Spring Boot, React, PostgreSQL, Redis*

- Formed a full-stack retail inventory management system using **Spring Boot and React**, managing **5,000+ products** with real-time stock tracking across multiple store locations.
- Orchestrated automated reorder and sales-reporting workflows backed by **Redis-based stock synchronization**, improving inventory availability accuracy by **25%** and supporting data-driven restocking decisions.

EDUCATION

Master of Science in Computer Science | New York Institute of Technology, New York, USA